

We now consider the ‘standard’ 3-dimensional objects in the study of multivariable calculus.

Given a 2-dimensional curve sitting in 3-space, say $y = x^2$ with $z = 0$, if we allow it to extend infinitely in the ‘missing’ direction, then the new object is called a **cylinder** generated by the 2-dimensional curve.

Most of the traditional 3-dimensional objects we study are extensions of 2-dimensional curves called *conic sections*. Some of these 2-dimensional objects are the following:

Circle: $x^2 + y^2 = r^2$

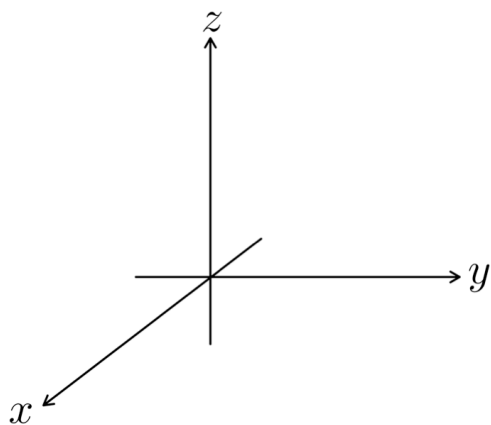
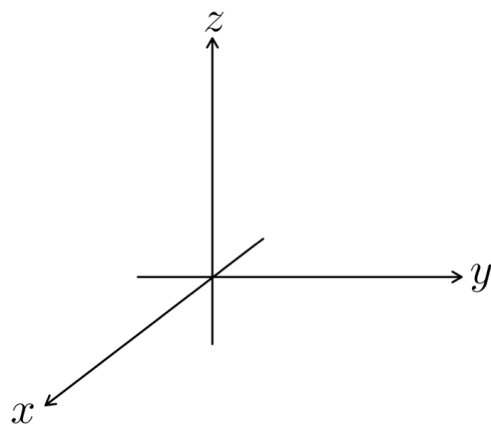
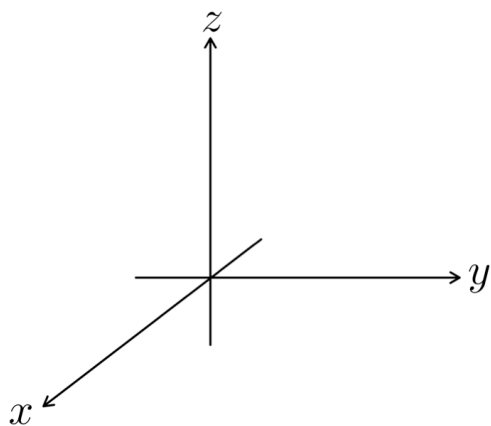
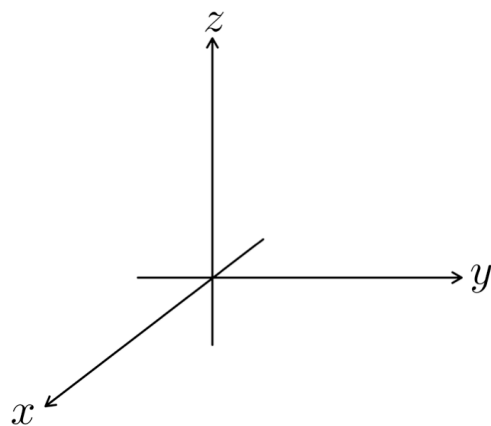
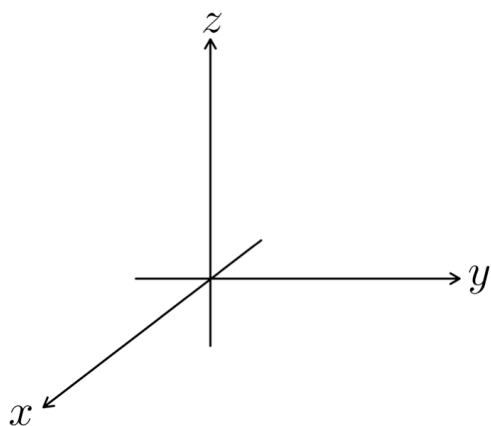
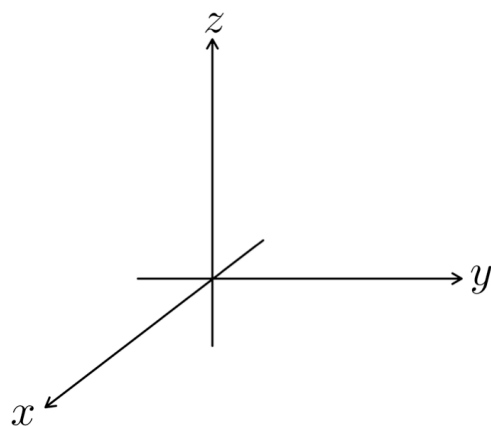
Ellipse: $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

Parabola: $y = ax^2$

Hyperbola: $\frac{y^2}{a^2} - \frac{x^2}{b^2} = 1$

When extending them to 3-space, the ‘correct’ way to view these new objects is to look at 2-dimensional slices of it, where these slices will look like the above shapes. You draw these slices in 2-space and then connect them to form the resulting 2-dimensional object. When you have an equation for the 3-space object, you can find an equation for a 2-space slice if you simply set one of the variables equal to a constant and then solve the equation.

Example. Consider the **hyperbolic paraboloid**: $\frac{y^2}{b^2} - \frac{x^2}{a^2} = \frac{z}{c}$, where $c > 0$

**Ellipsoid****Elliptical Paraboloid****Elliptical Cone****Hyperboloid of One Sheet****Hyperboloid of Two Sheets****Hyperbolic Paraboloid**